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Unit- 9:

STRESS, ADPTATION & COPING PATTERN

Objectives:-

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- Explore the biological, psychological, and social aspects of the human stress response.
- Describe the difference between adaptive and maladaptive coping strategies.
- Understand coping strategies that increase resilience to promote productive living and healthy aging.

Overview of Terminology

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- **Stress:** A state of disharmony or a threat to homeostasis
Physiological changes increase alertness, focus, and energy
Perceived demands may exceed the perceived resources
- **Coping:** The ability to maintain control, think rationally, and problem solve
- **Resilience:** Resistant quality that permits a person to recovery quickly and thrive in spite of adversity

Stress

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Stress is a state produced by a change in the environment that is perceived as challenging, threatening, or damaging to the person's dynamic balance or equilibrium. A stimulus or situation that causes stress is stressor.

A stressor: an internal or external event or situation that creates the potential for physiologic, emotional, cognitive, or behavioral changes in an individual.

It is generally believed that biological organisms require a certain amount of stress in order to maintain their wellbeing.

However, when stress occurs in quantities that the system can not handle, it produces pathological changes.

Stress

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□ Eustress

- Manageable Stress can lead to growth and enhanced competence

□ Distress

- Uncontrollable, prolonged, or overwhelming stress is destructive.

□ Acute Stress

- Immediate response to a threat or challenge

□ Chronic Stress

- Ongoing exposure to stress, may seem unrelenting

Causes of Stress

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□ External causes

- Family, work, economics, work, school, major life changes, unforeseen events, etc.

□ Internal causes

- Worry, uncertainty, fear, attitudes, unrealistic expectations, etc.

Sources of Clinical Stress

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For patients

- Uncertainty
- Fear, Pain
- Cost
- Lack of knowledge
- Risk for harm
- Unknown resources

For nurses

- Poor patient outcomes
- Risk of making an error
- Unfamiliar situations
- Excessive workload
- Inadequate resources

Classification of Stressor: General Classification

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1. Internal Stressor

- ▣ Originate within the person. **E.g.** fever, feeling of guilt, menopause, pregnancy...

2. External Stressor

- ▣ Originate outside the individual. **E.g.** marked change in environmental temperature, change in family or social role, peer rejection...

3. Developmental Stressor

- ▣ Occurs at a predictable time throughout life

4. Situational Stressor

- ▣ Unpredictable may occur any time during life

Types of stressors

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- **Stressors** exist in many forms and categories.
- They may be described as physical, physiologic, or psychosocial based on source.

According to source

- **Physical stressors** include cold, heat, and chemical agents;
- **Physiologic** stressors include pain and fatigue.
- E.g. of **psychosocial** stressors are fear of failing an examination and losing a job.
- Stressors can also occur as normal life transitions that require some adjustment, such as going from childhood into puberty, getting married, or giving birth.

According to involved persons

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1. **Day to day stressors:** as getting caught in a traffic jam, experiencing computer downtime, and having an argument with a spouse or roommate.
 - These less dramatic, frustrating, and irritating events—daily hassles—have been shown to have a greater health impact than major life events b/c of the cumulative effect they have over time.

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- 2. Major complex occurrences involving large groups, possibly even entire nations:** These include events of history, such as terrorism and war, w/c are threatening situations when experienced either directly, in the war zone, or indirectly, as through live news coverage
- The demographic, economic, and technological changes occurring in society also serve as stressors.

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3. Less frequent stressors involving few groups: relatively infrequent situations that directly affect the individual.

- This category includes the influence of life events such as death, birth, marriage, divorce, and retirement

The main categories of stress are:

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1. Frustration

- The emotional state induced when one is blocked from reaching a goal. Ex. *traffic jams*.

2. Pressure

- The emotional state induced when one is confronted by personal responsibilities
- Pressure involves expectations or demands that one behave in a certain way...pressure to perform or to comply.

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3. *Conflict*

- The emotional state induced when one is torn between two or more potential courses of action.
 - types of conflict have been studied extensively:
 - ***Approach-approach*** – when a person has a choice b/n 2 attractive goals,
 - ***Avoidance-avoidance*** – when a person has a choice between 2 unattractive goals (e.g., 1 exam or 25 page paper for final grade)
 - ***Approach-avoidance*** – when a choice must be made about whether to pursue a single goal that has both attractive and unattractive aspects... results in vacillation, or going back and forth...rats actually run up and down a ramp in this type conflict

4. **Change** → Life changes are any noticeable alterations in one's living circumstances that require readjustment.

Adaptation:

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- A process of change that occurs as a person responds to stress is *adaptation*.
- **Adaptation** is a constant, ongoing process that *requires a change in structure, function, or behavior* so that the person is better suited to the environment; it involves an interaction between the person and the environment.

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- An **adaptive response** occurs when a stimulus from the internal or external environment *threatens a person's well being*.
- Thus adaptation is an attempt to maintain functioning through *physiological and coping mechanisms*.
- **Coping** is the cognitive and behavioral efforts made to master, tolerate, or reduce external and internal demands and conflicts. It involves *solving a problem, making decisions, and relieving tension* created by stress.

Components of Human adaptation

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- **Physiological adaptation:** the body responds to stressor to maintain functioning compatible with survival
- **Psychological adaptation:** emotional adaptation involves using psychological coping mechanisms that defend the client in the presence of a stress.
- **Developmental adaptation:** includes cognitive development and education, perceptions of other people and the world in general

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- **Socio-cultural adaptation:** the social network consisting of a family members, co-workers and peers may provide formal or informal psychological support to assist a person adapting to stress.
- **Spiritual adaptation:** includes beliefs about supreme being, a feeling of oneness with nature and the world and a positive sense of life's meaning and purpose.

Human response to stress

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□ Emotional Responses

- Annoyance, anger, rage
- Apprehension, anxiety, fear
- Dejection, sadness, grief
- Positive emotions(*promoting creativity and flexibility in problem solving, facilitating the processing of important information about oneself, and reducing the adverse physiological effects of stress*).

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□ Behavioral Responses

- **Frustration-aggression hypothesis** → *striking out at others aggressively*
- **Coping** → *defensive coping (erecting defense mechanisms), or constructive coping (realistically appraising situations and confronting problems directly).*
- **Giving up and blaming oneself** (*learned helplessness – passive behavior produced by exposure to unavoidable aversive events*)
- **Self-indulgement** (*eating, drinking, smoking, shopping*)

Stages of the Stress Response

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- ***General Adaptation Syndrome*** of Hans Selye (1907-1982)
- ***Selye's Local adaptation Syndrome, and General Adaptation Syndrome***

1. Local adaptation syndrome(LAS):

- This syndrome includes the inflammatory response and repair processes that occur at the local site of tissue injury.
- The local adaptation syndrome occurs in small, topical injuries, such as contact dermatitis.
- If the local injury is severe enough, the general adaptation syndrome is activated as well

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2. ***General adaptation syndrome:***

- A nonspecific response to diverse, noxious stimuli, or major threats.
- The term emphasizes that the response is both generalized and adaptive, *i.e. aimed at neutralizing the threat and restoring the body to homeostasis.*

Stages of generalized adaptation syndrome

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- i. **The alarm reaction**
- ii. **Resistance, and**
- iii. **Exhaustion**

- **Alarm**—when one feels threatened
 - ▣ Activation of the fight or flight reaction
- **Resistance**—mobilization of resources to solve the problem
 - ▣ Continued stress causes adaptation
- **Exhaustion:** Adaptation fails and level of function decreases

i. The alarm reaction phase

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- *The reaction phase* comprises an *initial shock phase* and a subsequent *counter shock phase*.
- The *shock phase* exhibits autonomic excitability and increased adrenaline discharge, and gastro-intestinal ulcerations.
- *At this level heart rate increases, increased oxygen intake, increased blood flow to the muscle and greater mental alertness.*

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- These extensive hormonal activity prepares the person for *“fight-or-flight”* response
- The *counter shock phase* marks the stimulation of the adrenals through the adrenocorticotrophic hormone (increases blood glucose and retain sodium and water in the body) and thyroid stimulating hormone (increases body metabolic response).

ALARM: Activation of Hypothalamic Pituitary Adrenal Axis

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Catecholamines

- Increases cardiac output
- Dilates airways
- Dilates pupils
- Mobilizes glucose
- Causes vasomotor changes
- Decreases digestion
- Enhances coagulation

Cortical Steroids

- Elevates glucose
- Increases amino acids
- Increases NA resorption
- Increases extracellular fluid volume
- Inhibits histamine and bradykinin
- Suppresses the immune response

ii. The stage of resistance

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- If noxious stimulation continues, the organism enters the ***stage of resistance***.
- A stage in which an individual is ***coping with stressor***
- At this ***time adaptation may occur***, involving modification of the external and internal environments
- The amount of ***resistance varies among individuals***, depending on the level of physical functioning, coping abilities, and total number and intensity of stressors experienced

Resistance....

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- *In this stage, the symptoms of the alarm reaction disappear*, which seemingly indicates the organism's adaptation to the stressor
- When the body is successful in its fight against the threat, the **GAS** ends with the stage of resistance.
- Adaptation occurs: activation of the hypothalamic pituitary adrenal axis continues
- The stressor may be resolved: the body returns to homeostasis
- May progress to exhaustion: stress continues as resources are depleted

iii. The stage of exhaustion

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- If the aversive stimulation persists, resistance gives way to the *stage of exhaustion*.
- *A stage in which there is failure to adapt to a stressor, leading to physical, psychological and emotional problems*
- Physical symptoms of the alarm reaction may briefly reappear in a final effort by the body to survive
- The individual in the stage of exhaustion usually becomes *ill* and may *die* if assistance from outside source is not available

Stages of Exhaustion....

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- This stage can often be reversed by external source of adaptive energy, such as medication, blood transfusion, etc...
- Without outside intervention, the outcome of the stage of exhaustion is death
- Occurs when the demands of the stress exceeds the persons ability to adapt.
- Functioning declines
- May result in health problems; Physical symptoms, mental symptoms

Sign and symptoms of stress

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- Stress is manifested in a variety of ways physical, emotional, behavioral, and cognitive.
- **Physical or Physiological :-** increasing HR, BP, RR and blood glucose level; dryness of the mouth, perspiration, dilation of the pupil, breathing difficulty, and numbness.
- Decreased immune response etc...
- **Cognitive:** cognition span decreases, decision making difficult, memory loss, mental block.
- Decreased concentration, comprehension,

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- **Emotional:** anxiety, aggression, apathy, boredom, depression, emotional outbursts, frustration, guilt, irritability, bad temper, moodiness, low self esteem, loneliness.
- Fear, anxiety, depression, fatigue
- **Behavioral:** accident proneness; excessive eating or loss of appetite; drinking, smoking, drug taking; excitability; impulsive behavior; speech problems; nervous laughter; restlessness.
- Irritability, withdrawal, violence

Factors affecting stress response

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Genetics

- Genes control the stress response
 - ▣ Individuals have different responses to stress
- There is a genetic component to:
 - ▣ fearful behavior
 - ▣ anxiety disorders
 - ▣ Neurobiological response

Development

- Life experiences can affect a person's stress response
- Social support
 - ▣ Strong support is protective
- Early life stress
 - ▣ Increases stress reactivity as an adult

Coping & Resilience

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Coping

- Ability to control emotions
- Ability to perceive reality
- Ability to think rationally
- Ability to problem solve
- Culturally defined

Resilience

- The ability to bounce back
- The positive capacity to cope with stress
- Provides resistance to negative events
 - ▣ Hardiness,
 - ▣ Resourcefulness

Coping

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Adaptive Coping

- ▣ Contribute to resolution of the stress response

Maladaptive Coping

- ▣ Strategies that cause further problems

Active Coping

- ▣ Actively seeking resolution to the stress

Stress management

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- ❑ The essential stress management include:
 1. Deal with the problem and understand the problem in order to make an effort to alter the situation responsible for the problem
 2. Deal with the feelings
 3. Use social support
 4. Reduce the physiologic arousal of stress such as exercise to burn off excessive arousal and relaxation methods and use of message.

Promote Adaptive Coping

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- **Realistic expectations**

- Set realistic goals

- **Planning**

- Anticipate problems, have a backup plan

- **Reframing**

- Change the way you look at things

- **Relaxation**

- Learn relaxation techniques, take time-out for leisure

Promote Adaptive Coping

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- **Discuss the problem**
 - ▣ Utilize existing social supports to problem solve
- **Training:** repairs for stressful events
- **Nutrition:** Eat healthy, avoid skipping meals
- **Exercise**
 - ▣ Include regular exercise
- **Sleep**
 - ▣ Get adequate sleep—avoid fatigue

Avoid Maladaptive Coping

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- Blurring of boundaries
- Avoidance/withdrawal
- Negative attitude
- Anger outbursts
- Alcohol/Drugs
- Hopelessness
- Negative self-talk
- Resentment
(bitterness)
- Violence

Promote Resilience (Relaxation) Factors

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- Positive Role Models
- Optimism
- Humor
- Moral Compass
- Altruism
- Religion & Spirituality
- Social Support

Positive Role Models

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Transmit:

- Attitudes
- Values
- Skills
- Patterns of thoughts and behaviors

Optimism

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- Positive Beliefs
 - Associated with well being
 - Cognitive reframing
 - Positive thinking
 - Refute the negative thinking
 - Believe in a meaningful cause
- It is important to acknowledge relevant negative factors

Humor:

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- Ability to appreciate or express or inclination or whim (a sudden desire or change of mind)
- Highly effective
- Mature coping mechanism

"Another of the souls' weapons for the fight for self-preservation, it is well known that humor, more than anything else in the human makeup, can provide an aloofness(away & to windward(toward stressor)) and the ability to rise above any situation, even for a few seconds." *Viktor Frankl*

Moral Compass

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- Conduct a moral inventory

"Look not for any greater harm than this, destroying the trustworthy, self-respecting, well-behaved man with you." Epicetus

- Maintain your integrity

"Between stimulus and response there is a space. In that space is our power to choose our response. In our response lies our growth and our freedom." *Viktor Frankl*

Altruism:

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- Selfless concern for the well-being of others or benefits another at its own expense
- Unselfish regarding the welfare of others
- Believe in a meaningful cause
- Mutual cooperation
- Activates of the brain's reward center

Religion & Spirituality

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- Associated with psychological and physical well being;
- Guards against despair (complete loss or absence hope or loss or be without loss)
- Provides social support
- Provides positive role models
- Provides a positive mission

Social Support

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- Social support has a profound effect on life expectancy
- Patients have better outcomes with strong social support
- Isolation and poor social support are associated with a poor stress response
- Few hardy individuals "go it alone"

Signature Strengths

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- Recognize skills and talents
 - ▣ Inventory strengths
- Use your strengths and talents
 - ▣ Decide what works
- Actively cope
 - ▣ Apply concepts to enable active coping.

Summary

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- Stress is part of everyday life
 - ▣ It can promote growth and competency
 - ▣ If unrelenting or overwhelming it can cause adverse effects
- Adaptive coping enhances resilience
- Maladaptive coping causes additional problems
- Enhanced coping increases resilience while diminishing the adverse affects of stress, thus promoting health

Summary: Practical Tips

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- Set realistic expectations
- Exercise regularly
- Eat healthy
- Get adequate sleep
- Maintain a work-leisure balance
- Positive Reframing & optimism
- Enhance social support

The patient may exhibit some of the diseases of adaptation:

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- Angina
- Depression
- Dyspepsia
- Eating disorder
- Fatigue
- Headaches
- Hypertension
- Impotence
- Insomnia
- Low back pain
- Myocardial infarction
- Sexual dysfunction
- Ulcers



THANK YOU!!!